MALAV NAIK

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Visa Status: Stamp 1G — Eligible to work full-time until Dec 2027

PROFILE

Data professional in Dublin with about 10 years of experience in data across India and Ireland. Core stack spans SQL, SQL Server, SSIS, Azure SQL, Python, dbt, DuckDB and Power BI, including an on-premise SQL Server to Azure SQL migration that cut infrastructure costs by about 25%. Works closely with business users and non-technical stakeholders, defining KPIs and building dashboards and reports for them. Builds GenAI and AI automation systems with retrieval-augmented generation, knowledge graphs and large language models, from an MSc thesis on personalised learning to a fairness-checked content moderation model (ROC-AUC 0.957). Has handled problems ranging from multi-million-record financial data pipelines and public open data to platform integrity escalations for a global technology client.

EXPERIENCE

Escalation Operations Analyst Mar 2026 – Present

Accenture – for a global technology client (Technology & Platform Operations, platform integrity), Dublin, Ireland

- Analyse complex escalations with AI-assisted tools every day, including LLM-driven case analysis and automated classification.
- Apply decision-tree logic to classify and resolve cases across 15+ categories.
- Coordinate 30+ concurrent cases across 6 business functions.
- Run process-gap analysis across 3 global operations centres.
- Keep audit-standard documentation.

Data Analyst & ETL Developer Feb 2019 – Aug 2024

S&P Global Market Intelligence, India

- Built SSIS pipelines over multi-million-record financial datasets.
- Designed data models for analytics.
- Helped migrate on-premise SQL Server databases to Azure SQL Managed Instance, cutting infrastructure costs by about 25%.
- Supported and troubleshot SQL Server and Azure SQL databases, monitoring data quality and performance.
- Built Power BI dashboards for non-technical users.

Data Analyst Oct 2015 – Jan 2019

Bright Info Solutions, Ahmedabad, India

- Pulled data with SQL, cleaned it in Excel and Python, and built weekly reports.
- Built dashboards and automated scorecards, defining KPIs alongside business users.
- Carried out exploratory data analysis.

KEY PROJECTS

- **Content moderation classifier, with a fairness check:** Trained a TF-IDF baseline and DistilBERT on 1.8 million public comments, set two threshold policies, simulated a review queue and checked false-positive rates by identity group. ROC-AUC 0.957; 36.6% of toxic comments auto-actioned at ≥ 90% precision; harmless comments wrongly flagged cut from 12.8% to 9.6%. Stack: Python, scikit-learn, PyTorch, DistilBERT, Kaggle, Hugging Face, ONNX, transformers.js.

- **Dublin bikes + weather data pipeline:** A collector every 15 minutes, then a daily dbt + DuckDB run into an incremental star schema with 14 automatic data tests and Telegram alerts. Every station, every 15 minutes, modelled and tested into a live dashboard and public dataset, for €0. Stack: Python, GitHub Actions, dbt, DuckDB, Parquet, Hugging Face Datasets, GBFS, Met Éireann.
- **Ask the EU Digital Services Act:** Split the official text into 482 chunks and compared keyword, semantic and hybrid search on 25 hand-written questions; the demo quotes the law only. The right article is in the top 3 for 84% of test questions (72% in the free browser version). Stack: Python, BeautifulSoup, sentence-transformers, BM25, transformers.js, Kaggle, Hugging Face, GitHub Actions.
- **SkillBot: from job title to the skills it needs:** MSc research on an LLM plus a knowledge graph, and a free in-browser rebuild that matches a job title to similar public job ads. About 5 of 10 suggested skills appear in the real ad, in a leave-one-out test on 1,167 ads (48%). Stack: Python, LLMs, Knowledge graphs, Prompt engineering, sentence-transformers, transformers.js, Hugging Face, GitHub Actions.
- **Irish rents, 2008–2025: an open dataset, cleaned:** A reproducible Python cleaning pipeline with a Kaggle notebook that rebuilds the data from the official source. 201,912 rents for 437 places, 2008–2025, published on Kaggle and Hugging Face. Stack: Python, pandas, Kaggle, Hugging Face Datasets, GitHub Actions, CSO open data.
- **Blog automation bot with Telegram approval:** A Python bot on GitHub Actions that previews each post on Telegram and publishes only after one tap. Posts go to the blog, DEV and LinkedIn in about a minute of my time, on GitHub's free tier. Stack: Python, GitHub Actions, Telegram Bot API, DEV (Forem) API, LinkedIn Posts API, Jekyll.
- **Gmail auto-reply with Telegram approval:** Claude drafts a reply as a Gmail draft and sends it to Telegram with Send and Discard buttons. A routine reply becomes one tap, and the AI never sends anything by itself. Stack: Python, Claude API, Gmail API, Telegram Bot API, Prompt engineering.
- **Employment Permits 2025 dashboard:** Cleaned the official 2025 permits dataset, grouped companies into sectors and built a single-page Chart.js dashboard. A searchable, serverless dashboard of permits by sector, month and employer. Stack: Excel, Data cleaning, Chart.js, JavaScript, Tailwind CSS.
- **Hybrid Data Warehouse & Visualisation Application (MSc):** Built ETL across SQL Server, PostgreSQL and MongoDB over multi-source datasets with millions of records, with dashboards for non-technical stakeholders.

All projects, with code and sources for every number: malavnaik.com/work/

SKILLS

- **Data engineering:** SQL, SQL Server, SSIS (ETL), Azure SQL, Data modelling, dbt, DuckDB, Parquet, GitHub Actions
- **Analytics & BI:** Python, pandas, Power BI, Excel, Chart.js, Kaggle
- **GenAI & ML:** LLMs, RAG, Knowledge graphs, Prompt engineering, Claude API, sentence-transformers, BM25, scikit-learn, PyTorch, DistilBERT, ONNX, transformers.js, Hugging Face
- **Automation & web:** Telegram Bot API, Gmail API, DEV API, LinkedIn API, Jekyll, JavaScript, Tailwind CSS, BeautifulSoup
- **Cloud:** Azure PaaS Databases, Azure SQL MI, AWS QuickSight, AWS Glue
- **Operations:** Platform Integrity Analysis, Escalation Decision Frameworks, Cross-Functional Workflow Orchestration, Incident Classification, SLA Management, Process Gap Analysis, Compliance Documentation

EDUCATION

MSc Data Analytics Sept 2024 – Sept 2025

National College of Ireland, Dublin, Ireland

- **Modules:** Data Mining, Machine Learning, Data Visualization, Optimization, Deep Learning, Generative AI
- **Dissertation:** *Enhanced Personalized Learning Experiences by Leveraging Knowledge Graphs and Prompt Engineering*

Bachelor of Computer Applications 2015

Veer Narmad South Gujarat University, Gujarat, India

CERTIFICATIONS

- Build Your Generative AI Productivity Skills with Microsoft and LinkedIn
- Google Analytics Certification
- LLM Tech Essentials (Levels 1–3)